

A SHORT-TERM TEMPERATURE FORECASTER BASED ON A NOVEL RADIAL BASIS FUNCTIONS NEURAL NETWORK

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Abstract

Many applications dealing with electric load forecasting in buildings require temperature prediction. A new method for short-term temperature forecasting based on a Radial Basis Functions Neural Network, initialized by a Regression Tree, is presented. In this method, each terminal node of the tree contributes one hidden unit to the RBF network. The forecaster uses the current coded hour and the temperature as inputs, and predicts the next hour temperature. The results demonstrate this predictor can be used for load forecasting.

1. Introduction.

As it is widely shown in the scientific literature, the vast majority of the Short Term Electric Load Forecasters (STLF's) use the environment temperature as the main weather variable (Ref. 1,2,3,4). Many interesting results compute the electric consumption as a relationship between the external temperature value and the load consumption value. In Figure 1, the normalized graphic is an example of this closed interaction [the load variables (solid curve) were measured by an Energy Management System at Sevilla Hotel, Havana, a building with whole power consumption about 1 MW].

However, in a practical application for load prediction, the temperature must be predicted too. This implies that the first task required in Load Predictor Systems, where the environmental temperature is an important input variable, is the on-line temperature prediction. STLF can be designed for local, regional or national predictions. In this paper, local refers to a building area. Hence, the temperature prediction method tested here is for a STLF in buildings.

Many methods for short-term temperature prediction have been developed. An interesting method for temperature prediction was proposed in (Ref. 5). The hourly temperature forecasting is obtained by stretching or contracting the daily profiles to match the known daily high and low values. Consequently, the predicted high and low temperature values are needed. In (Ref. 6), an ANN temperature forecaster was developed to provide hourly temperature forecast for up to seven days in advance. The daily forecast is based on the most recent hourly temperature and the predicted high and low temperature value of the day to be forecasted. The authors recognize that these two values have to be supplied, for instance, by a National or Local Weather Service. In fact, the high and low temperature values provide an indication of future trends (such as a cold front, etc.), but in a real-time application, it is not easy or practical to supply these values every day. For this reason, in our application, the predictor has been designed without using the daily high and low temperature values.

The method proposed here, similarly to other ideas, makes use of the computerized systems and automatic equipment installed at those buildings where a STLF system is applied. This means to take advantage of the presumably included environmental temperature measurement, so, additional technology

is not required, it is only needed to store a database with past atmospheric temperature values and make use of this database for studying the dynamic behavior of temperature and thus, forecast its future values.

Generally, STLFs make hourly load prediction. Hence, the temperature predictor proposed here must provide temperature in this same period.

This new short-term temperature forecaster is based on a combination of Regression Trees and Radial Basis Function Neural Networks. Section 2 presents theoretical support of the predictor. In Section 3, the temperature predictor is discussed. Section 4 shows results and comparisons, and finally, conclusions are reported in Section 5.

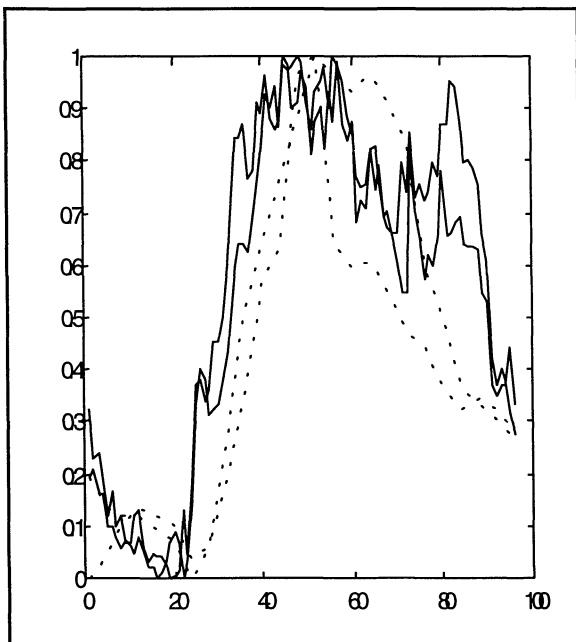


Figure 1. Load (solid) and Temperature (dashed) relationship.

2. Theoretical Support.

2.1. Regression Trees and RBF Neural Networks.

2.1.1. Radial Basis Function Neural Networks. (RBFNN).

Radial Basis Function (RBF) Neural Networks (Ref 7) transform a non-linear n -dimensional input space to an m -dimensional space and then estimate a model using linear regression. The non-linear

transformation is controlled by a set of m basis functions each characterized by a position or center $\mathbf{c}_j$ in the (original) input space and a width or radius vector $\mathbf{r}_j$, $j \in \{1, 2, \dots, m\}$. The basis functions are usually local in the sense that they respond most strongly to the inputs nearest to the center $\mathbf{c}_j$ in the metric determined by the radius $\mathbf{r}_j$. Typically, they are Gaussian-like functions, but other classes are also frequently used.

Algorithms for building RBF networks often operate in two phases. The first phase selects the center locations and radii, and the second, estimates the weights $\mathbf{w}$, of the linear regression model.

$$f(x) = \sum_{j=1}^m w_j h_j(x) \quad (1)$$

For example, the first phase could simply allocate an RBF center to each input datum in the training set. Without further modification, this scheme would normally produce a complex model, which would overfit the data. However, if the weights for the linear model are subsequently calculated using penalized likelihood, the model complexity can be controlled by one or more regularization parameters whose values can be determined by optimizing a suitable model selection criterion. When the training set comprises many data, a heuristic that randomly selects a subset of input data vectors as centers makes sense and also avoids a heavy computational burden when performing the matrix inversion necessary to calculate the weights.

In an alternative approach, initially, each input can contribute one candidate RBF and a subset selection procedure can then choose which to include in the model. The approaches of (Ref. 8) and (Ref. 9) fall into this category. In both cases, forward selection was used, where the network is initially small and is subsequently built up to its optimal size (Ref 10).

2.1.2 Regression Trees. (RT)

The basic idea of a regression tree is to split the input space recursively in two and approximate the function in each half by the average output value of the samples it contains. The root node of the tree is the smallest hyperrectangle that will include all the training data, $\{\mathbf{x}\}_{i=1}^p$. Its size s_k and center c_k for each dimension k are:

$$s_k = \frac{1}{2} (\underbrace{\max(x_{ik})}_{i \in S} - \underbrace{\min(x_{ik})}_{i \in S}) \quad (2)$$

$$c_k = \frac{1}{2} \left(\underbrace{\max(x_{ik})}_{i \in S_L} + \underbrace{\min(x_{ik})}_{i \in S_R} \right) \quad (3)$$

where $S = \{1, 2, \dots, p\}$ is the set of training set indices. A bifurcation of the root node divides training samples into left and right subsets, S_L and S_R , on either side of a boundary b of the dimensions k such that:

$$S_L = \{i: x_{ik} \leq b\},$$

$$S_R = \{i: x_{ik} > b\}.$$

The mean output value on either side of the bifurcation is

$$\bar{y}_L = \frac{1}{p_L} \sum_{i \in S_L} y_i, \quad (4)$$

$$\bar{y}_R = \frac{1}{p_R} \sum_{i \in S_R} y_i \quad (5)$$

where p_L and p_R are the number of samples in each subset. The residual square error between model and data is then:

$$E(k, b) = \frac{1}{p} \left(\sum_{i \in S_L} (y_i - \bar{y}_L)^2 + \sum_{i \in S_R} (y_i - \bar{y}_R)^2 \right) \quad (6)$$

The bifurcation which minimizes $E(k, b)$ over all possible choices of k and b is used to create “children” of the root node and is found by a simple discrete search over n dimensions and p cases. The children of the root node are split recursively in the same manner and the process terminates when all remaining bifurcations create children containing fewer than $p_{\min}$ samples, where $p_{\min}$ is a variable parameter. The child centers will be shifted with respect to their parent nodes and their sizes reduced in dimension k .

2.2. Transforming Tree Nodes into RBFs.

The regression tree contains a root node, some non-terminal nodes (having children) and some terminal nodes (having no children). Each node is associated with a hyperrectangle of input space having a center c and size s as described above. The combination of regression trees and RBF for non-parametric regression was suggested by (Ref. 11) and improved by (Ref. 12). To transform a hyperrectangle into a Gaussian RBF, centers c are used as the RBF centers

and the sizes s , scaled by a parameter α , are the RBF radii, so $r = \alpha s$.

A new way for selecting the nodes of the network from the generated tree was proposed in (Ref. 10) where the searching for RBF candidates is made by traversing the tree from the largest hyperrectangles at the root to the smallest ones at the terminal node, including the children nodes. Those nodes, which most decrease the model selection criterion, are selected.

After the centers are selected, the weights w are computed by solving the equation:

$$w = (H^T H)^{-1} H^T y \quad (7)$$

where H is the design matrix (its elements $h_j x(i)$ are computed by the equation which defines the basis function used), and y is the output training set.

Figure 2 illustrates the effectiveness of combining RBFNN and RT. The example is taken from (Ref. 10) and shows how a neural network trained by this method permits to reconstruct a function previously corrupted by a normally distributed noise with variance equal to 0.05. The following polynomial function is evaluated with 100 data in the interval $[-4, 4]$.

$$f(x) = 1 + (1 - x + 2 * x^2) * e^{-x^2} \quad (8)$$

3. The short-term temperature forecaster based on the novel RBF networks.

As it is obvious, many environmental conditions can affect the outside temperature: seasons, hour, wind, solar intensity, rain, etc. Because of the approach proposed here is focused in predicting the temperature value on very small areas like buildings or similar, only a hypothetical closed system has been considered. The affecting factors on this system, which we consider can influence over the dynamic behavior of the environment temperature, are: the hour and the previous temperature value with respect to the instant of prediction. With this consideration, no large databases are needed for training the neural network.

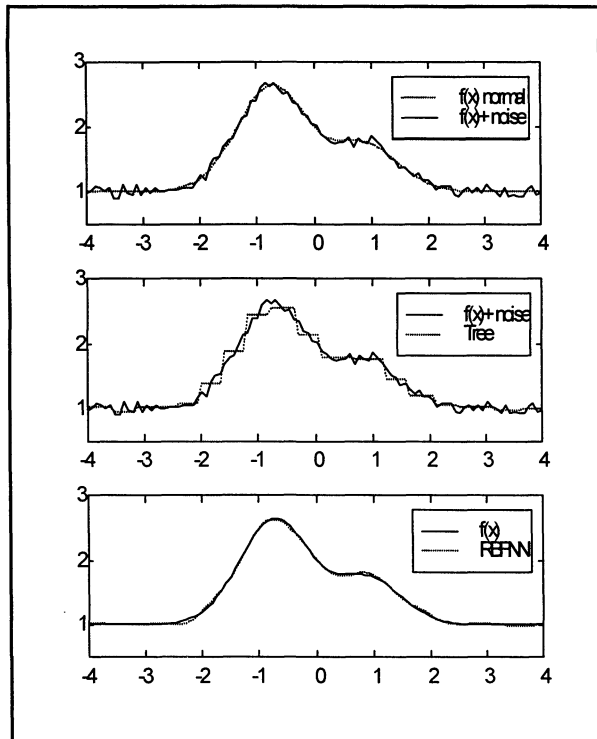


Figure 2. Example of the combination between RBFNN and RT.

The training data are formed with a moving window, following the strategy used in (Ref. 6). By means of this mode, the inherent non-stationary behavior of the outside temperature is reflected in the daily updating of the weights. In this paper, we have found that a window of 28 days has a good size, but it could be case depending.

The following table shows how the training set is formed.

Table 1. Training set.

Variable	Data
Input Temperature	$T_{h=0}^{d-n} \dots T_{h=23}^d$
Hour	$\sin(\pi * ((h_0^{d-n} \dots h_{23}^d)/12))$ $\cos(\pi * ((h_0^{d-n} \dots h_{23}^d)/12))$
Output Temperature	$T_{h=1}^{d-n} \dots T_{h=0}^{d+1}$

Where:

$T_{h=0}^{d-n} \dots T_{h=23}^d$: Set of temperature values from the hour (h) 00:00 of the day when the window begins (d-n days) until the

hour 23:00 of the day (d) when the training is carried out.

$\sin(\pi * ((h_0^{d-n} \dots h_{23}^d)/12))$; $\cos(\pi * ((h_0^{d-n} \dots h_{23}^d)/12))$

: Code of the hours. (00:00 to 23:00).

$T_{h=1}^{d-n} \dots T_{h=0}^{d+1}$: Set of temperature values from the hour (h) 01:00 of the day when the data window starts (d-n days) until the hour 00:00 of the day (d+1) when the training is carried out.

The resultant input database is composed by two elements: the hour and the last value of temperature. In that manner, the use (or necessity) of high and low temperature values is avoided. As an extended practice for coding the hours, (Ref. 13, 14) their sine and cosine values have been used. The output for training is the next outside temperature.

3.1. Neural Networks parameterizations.

Several parameters have to be configured for using the model developed by (Ref. 10). In order to perform this work, his very useful MATLAB based functions for RBF^a were used.

After some exhaustive experimentation, the best results were obtained configuring the predictor as it is shown in Table 2. Of course, this configuration is case depending too, so, a previous analysis, in order to choose the right parameters, should be done before applying the predictor to a real time operation.

Table 2. Predictor Configuration.

Parameter	Description	Value
p_{min}	controls the depth of the regression tree	3-5
α	determines the relative size between hyperrectangles and RFB centres	10
msc	the model selection criterion	BIC (Bayesian Information Criterion)
type	type of RBF function	multi-quadratic

Re-training of this neural network is performed daily with the last moving window. If the chosen length of

^a Available at www.anc.ac.uk/~mjo/software/rbf2.zip

these windows is, for instance, 28 days, the input data set forms a matrix with three rows, T_i , $\sin(h_i)$ and $\cos(h_i)$; and 672 columns (24 hours*28 days). For a standard RBF neural network, it could be quite difficult to select the right set of centers. However, the selection of the RBF's centers by regression trees, not only produces an accurate model, but also with a reduced number of centers. In the experiment explained below, an average of less than 20 centers was obtained.

As a validation criterion for the quality of this daily training, computing an error criterion, the Mean Absolute Error (MAE) is proposed:

$$\varepsilon = \frac{1}{n} \sum_{k=1}^n |e(k)|; \quad e(k) = Tp(k) - Ta(k) \quad (9)$$

[where n is the total number of hours, ε is the MAE, $e(k)$ is the error between the forecast temperature $Tp(k)$, and the current temperature $Ta(k)$, at hour (k)], and check if it is lower than one previously established. It should be remembered that the main objective of this forecaster is for STLF systems, but the electric load consumption really depends on more factors. A good consideration (at least, for load prediction) is to get a MAE around 0.5 °C (in Ref. 6, the result reported, 0.82 °C, was the average of MAE in ten utilities).

4. Results.

In order to determine how well this method works, data from the Great Energy Predictor Shootout II, organized by ASHRAE (Ref. 15) were used. This temperature data set is from central Texas, and from the period May 20th to August 20th. Data for input have been normalized using the following equation:

$$X_n = 2 * \left[\frac{X - X_{min}}{X_{max} - X_{min}} \right] - 1 \quad (10)$$

Thus, new normalized data between $[-1,1]$ were obtained. Taking into account that the weights are updated daily, X_{min} and X_{max} are selected choosing, in the case of the temperature, the range of maximum and minimum Celsius degrees known for the zone. It is also possible to add some Celsius degrees to the respective high and low values occurred during the window of days.

Figures 3, 4 and 5 show the behavior of this new predictor during several weeks. As it can be

appreciated, if the environmental temperature behaves in a stable way, the results are very good (see Figure 3).

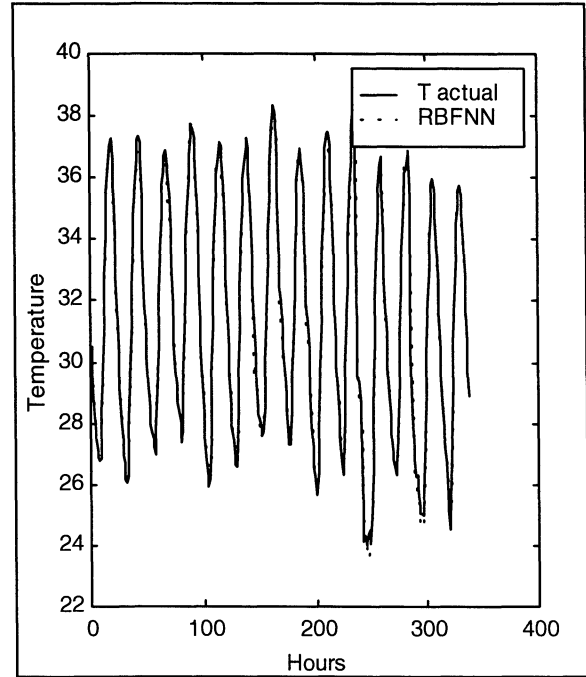


Figure 3. First two weeks.

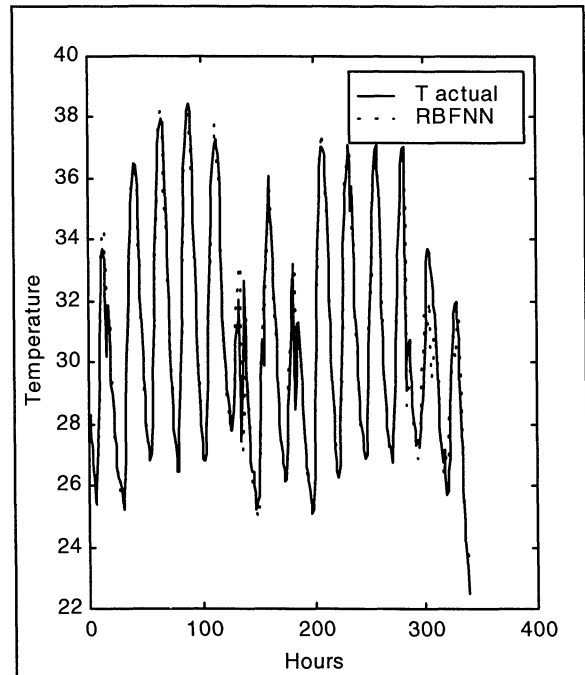


Figure 4. Third and four weeks.

If Figure 4 is carefully analyzed, some rare behaviors of the temperature can be observed. It could be supposed that they are wrong values of the data set, however, in order to test the forecaster in these conditions, these values were not eliminated. Fortunately, the predictor was able to follow these abrupt changes with a small error.

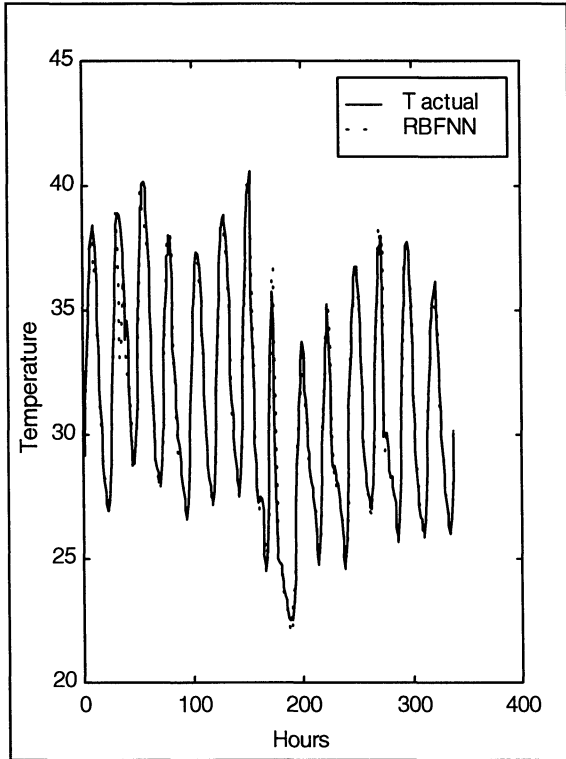


Figure 5. Last weeks.

4.1. Comparison with an ARX Model.

In order to compare this predictor with a linear model, a linear *AutoRegressive with eXogenous inputs (ARX Model)* was also implemented as forecaster with the same database. This model is mathematically expressed as:

$$A(q^{-1})y(k) = B(q^{-1})u(k) + e(k) \quad (11)$$

where $y(t)$ is the output model; $u(t)$, input model and $e(t)$, the error model; A, and B are the model parameters, and q^{-1} , the shift back operator.

For a first order model, the ARX predictor yields:

$$\hat{y}(k) = b_0 u(k-1) - a_1 y(k-1) \quad (12)$$

Since b_0 is composed by two elements (sine and cosine of the hour), the ARX Temperature predictor can be expressed as:

$$\hat{T}(k) = b_0 \cos(\text{hour}(k-1)) + b'_0 \sin(\text{hour}(k-1)) - a_1 T(k-1) \quad (13)$$

By means of the MATLAB Identification Toolbox, it was found an average MAE for this type of predictor (analyzing the same 90 days) equal to 0.5247 °C. The average MAE for the same period, using the predictor designed here, is equal to 0.4466 °C. Figure 6 shows the difference in MAE obtained by both predictors and how the RBF based predictor outperforms the linear one.

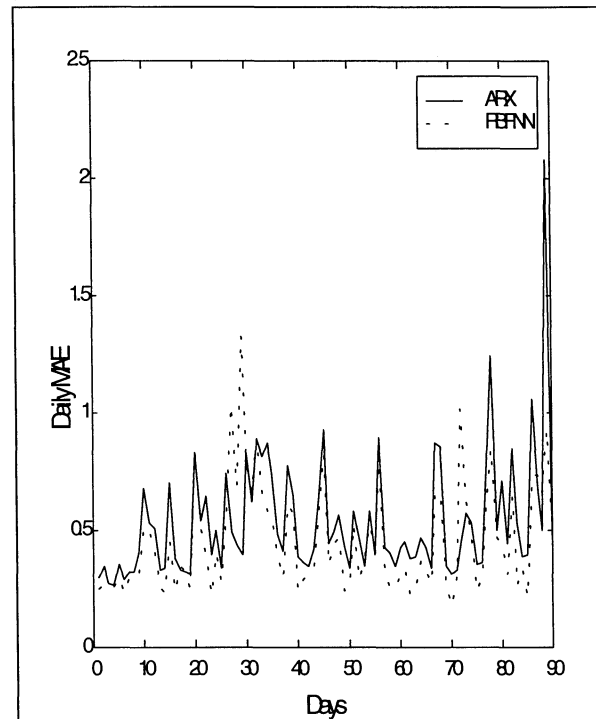


Figure 6. MAE for the RBFNN and the ARX based predictors.

5. Conclusions

In this paper, a new approach for short-term environmental temperature prediction, based on a novel RBF neural network, has been developed. The predictor is proposed for supplying the temperature when this is the main variable input in a Short Term Load Forecasting System. The results show how the proposed forecaster successfully models the non-linear behavior of the environmental temperature without using predicted high and low value of this variable for scaling or adjustment.

One of the main advantages of this method is that the final number of Radial Basis Function centres is small in relation to the whole input data. Besides, from a set of p_{mim} , and α parameters, the algorithm automatically chooses the best ones that models better the input data set. Hence, the chosen neural network facilitates the training (and re-training) and consequently, a shorter database can be used.

Although results are highly encouraging, more research should be done to improve the behavior of the predictor when extreme changes in atmospheric conditions take place, mainly when a decreasing of temperature suddenly occurs.

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